Brandon Miner

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AI Engineer — Multi-modal pipelines, LLM-assisted labeling, and production inference for unstructured video and text

Education

University of San Francisco — **San Francisco, CA**
M.S. in Data Science and Artificial Intelligence — *Expected Jun 2026*

- *Distributed Computing*
- *Machine Learning*
- *Deep Learning*
- Mlops

San Diego State University — **San Diego, CA**
B.S. in Statistics with Emphasis in Data Science, Minor in Mathematics — *Dec 2025*

Experience

AI/ML Engineer — **San Francisco, CA**
ACLU — *Oct 2025 – Present*

- Architecting the software design of an end-to-end **multi-modal AI** video auditing platform from scratch (**Python, FastAPI, MySQL, Docker**) to automate analysis of unstructured police bodycam footage, projected to eliminate **300+ hours** of manual review per monthly cycle, all while tracking product development.
- Engineered a **GPU-accelerated inference pipeline** using **FFmpeg** and **OpenAI Whisper** for speech-to-text transcription, processing **600+ videos per month**; integrated **LLM-driven auto-labeling** to generate training data for fine-tuning multi-stage text classification models.
- Architected and shipped a **multi-stage NLP classification system** combining LLM-assisted auto-labeling, iterative fine-tuning, and structured output parsing — delivering audit-ready signals directly in the hands of end-users.
- Designed and deployed a **private-server FastAPI backend** paired with a **React.js** frontend, delivering the full AI auditing system as a production-ready web application for non-technical end-users.

Data Scientist, Lead Intern — **San Diego, CA**
San Diego County Taxpayers Association — *Mar 2024 – Aug 2025*

- Led a team of **4 analysts** to build data pipelines delivering statistical analysis of civil workforce compensation and homelessness intervention programs.
- Developed an **data standardization pipeline** combining iterative web scraping and text normalization to unify messy public wage records; applied regression modeling to quantify a **19.7% police wage increase** per **1%** staffing reduction.
- Applied **mixed-effects regression models** and **multicollinearity reduction** to analyze regional homelessness expenditure data, identifying high-impact interventions (e.g., rapid rehousing) to guide future resource allocation decisions.

Technical Projects

GenAI Franchise Personality Quiz

- Built an end-to-end quiz generator: users describe any fictional class system in natural language; Gemini parses the franchise, scrapes Fandom wikis for canon characters, and roleplays each through 15 Likert questions to produce reference vectors asynchronously via FastAPI background jobs.
- Implemented a vector-based classifier that scores user answers with cosine similarity against LLM-generated character embeddings, ranks classes by per-type mean similarity, and projects results into 3D PCA for interactive visualization in a Next.js results UI.
- Deployed a split architecture on Vercel (frontend) and Google Cloud Run (API), with durable quiz artifacts in GCS, shareable quiz URLs, a community quiz library, and GitHub Actions CI (pre-commit, pytest, automated API deploy).

California Grape ETL Pipeline & Dashboard

- Engineered a **multi-source ETL pipeline** in Python to extract, transform, and consolidate disparate agricultural datasets into a unified schema ready for downstream ML and analytics workloads.
- **Containerized the full-stack application** using Docker Compose and deployed a serverless instance to **Google Cloud Run**, ensuring scalable and reliable cloud hosting.

- Built an interactive **Streamlit front-end** dashboard for dynamic visualization and filtering of regional production metrics; demonstrates end-to-end deployment from pipeline to user-facing interface.

Parkcast-SF: San Francisco Parking Assistant

- Owned system design and the production **FastAPI** inference API for a full-stack USF capstone parking platform, serving block-level **LightGBM** occupancy predictions from artifacts in **GCS**.
- Applied **K-nearest neighbors**-derived baselines to support **LightGBM** inference on sparse, non-metered blocks where direct sensor history was unavailable.
- Built automated model **CI/CD** on **GCP**: scheduled **GitHub Actions** retraining with **MLflow** experiment tracking and a **champion/challenger** promotion gate (validation MAE) before GCS upload; split deploy across **Vercel** (frontend) and **Cloud Run** (API).

Credit Card Fraud Detection

- Built a binary classification model using **Logistic Regression** (**Scikit-Learn**) with **PCA**-based dimensionality reduction to isolate the 16 highest-variance features for fraud detection.
- Performed **threshold optimization** via ROC-AUC analysis to minimize False Negative Rate while strictly capping False Positive Rate at **under 2%**, directly maximizing business value.

Technical Skills

Programming & Frameworks: Python (PyTorch, Transformers, Hugging Face, FastAPI, Pandas, NumPy, Scikit-learn), SQL (PostgreSQL), JavaScript (React.js), Bash
AI / ML: LLM integration & prompting, RAG/agent orchestration, multi-modal AI, NLP (ASR, Whisper, text classification, fine-tuning), GPU-accelerated inference, ROC-AUC threshold optimization, PCA
MLOps & Deployment: Docker, Docker Compose, Google Cloud Run, REST APIs (FastAPI), private-server deployment, GitHub Actions CI/CD, Git
Data: multi-source ETL pipelines, web scraping, entity standardization